

Jeffrey Xu Design Portfolio

Seeking a Fall 2026 co-op

Electrical Engineering student at the University of Waterloo, interested in hardware and power.

BACKGROUND

About

I'm an Electrical Engineering student at the University of Waterloo (Honours, 3.72 GPA), drawn to hardware and power. I currently design electrical hardware for UW Orbital, Waterloo's satellite design team, where I built an EPS current-sense circuit around an INA180 amplifier. I've also worked as a research and QA engineering intern at Groupthink Labs. I like projects that run the full stack — from schematic capture and PCB layout in Altium, to firmware in C on Arduino, to digital logic in Verilog on FPGAs.

Honours Electrical Engineering	University of Waterloo	GPA 3.72
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TOOLKIT

Skills

Hardware & Lab

Altium Designer	FPGA	Analog & Digital Circuits
Circuit Analysis	Component Selection	Oscilloscope
Digital Multimeter (DMM)	Soldering	Arduino
Raspberry Pi		

Software & Firmware

C	C++	Python
Verilog	Quartus Prime	SPI

I2C	VS Code	GitHub
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SELECTED WORK

Projects

Portable Sensory Cube

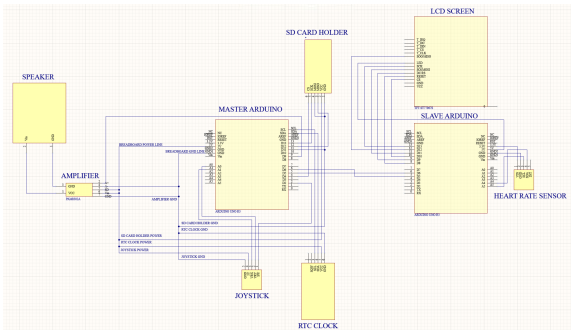
ARDUINO MEDICAL ASSISTIVE DEVICE

An Arduino-based assistive device aimed at mitigating hospital-induced delirium. It drives a TFT LCD over SPI to guide breathing exercises, and reads a MAX30102 pulse oximeter over I2C — configuring the sensor's registers to sample real-time photoplethysmography (PPG) data for heart-rate processing. Housed in a 3D-printed chassis with a joystick, speaker, and RTC clock.

[Arduino] [C] [SPI] [I2C] [MAX30102]



finished device



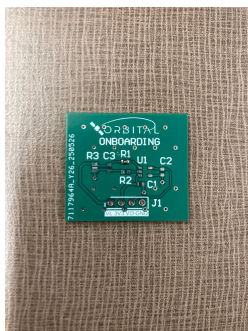
wiring schematic

IC Current-Sense Amplifier

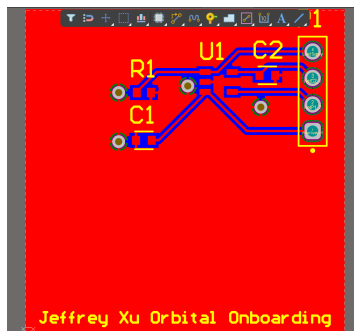
UW ORBITAL / SATELLITE EPS HARDWARE

A current-sense circuit for UW Orbital's satellite power system, built around an INA180 amplifier and a 0.22 Ω shunt resistor sized to limit power loss. I followed the INA180 datasheet layout guidance — checking capacitor voltage ratings and shunt power rating — to improve measurement stability and cut noise in the sense path, then laid out and soldered the PCB in Altium Designer.

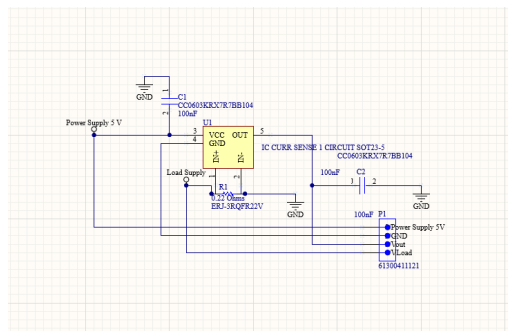
[Altium Designer] [INA180] [Analog Design] [PCB Layout] [Soldering]



soldered PCB



Altium layout



schematic

GET IN TOUCH

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